HEALTHCARE ANALYSIS PROJECT

BY TEAM – 2

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INTRODUCTION

- Healthcare analysis is the process of examining and evaluating data related to healthcare systems, services, policies, and outcomes to gain insights and inform decision-making.
- It involves analyzing various aspects of healthcare, including clinical data, financial data, operational data, and patient data, to identify patterns, trends, and correlations that can be used to improve healthcare delivery, patient outcomes, and policy decisions.
- The goals of healthcare analysis can vary depending on the context, but generally include improving patient care, enhancing operational efficiency, optimizing resource allocation, identifying and managing health risks, evaluating interventions and policies, and supporting evidence-based decision-making.
- Healthcare analysis plays a critical role in driving evidence-based practice and policy in healthcare, enabling healthcare professionals, policymakers, and administrators to make informed decisions to improve patient outcomes and population health.

KPIS SELECTED

- Number of Patients across various summaries
- Profit Vs Non-Profit Stats
- Chain Organizations w.r.t. Total Performance Score as No Score
- Dialysis Stations Stats
- # of Category Text - As Expected
- Average Payment Reduction Rate

IMPLEMENTED METHODOLOGY

Data Cleaning :

- Used Excel, Power BI, and Tableau to clean the data and ensure its quality.
- Removed any duplicate or irrelevant data to make the data set more manageable.
- Handled any missing data by using mean imputation for numerical data and mode imputation for categorical data.
- Checked for any errors, inconsistencies, or formatting issues and resolved them.

Data Visualization:

- Created visualizations using Excel, Power BI, and Tableau to explore the data and identify patterns and trends.
- Used various visualization types, such as scatterplots, histograms, and bar charts, to highlight different aspects of the data.

- Used a combination of static and interactive visualizations to engage the audience and promote understanding of the data.

- Used color, size, and other visual cues to highlight key patterns and trends in the data.

- Provided an interpretation of each visualization, highlighting key insights and any significant observations.

Data Analysis:

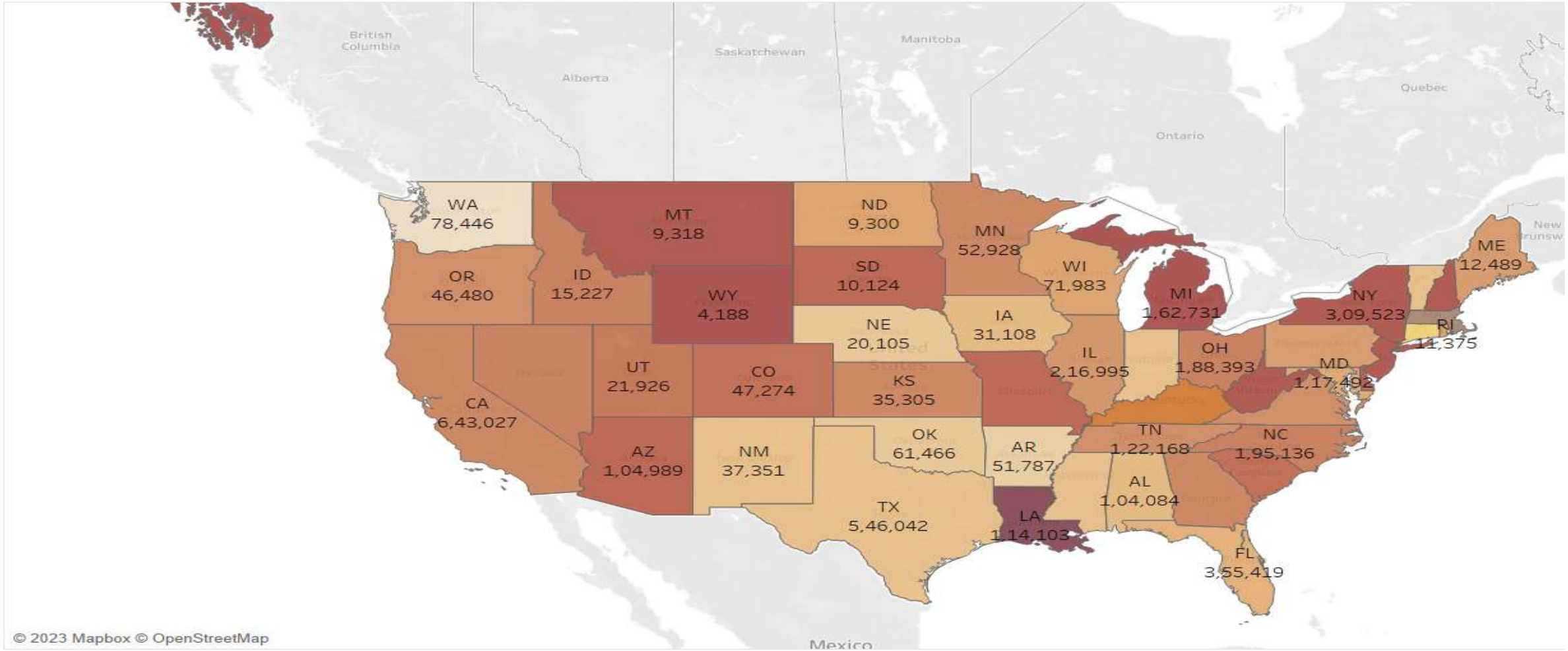
- Conducted descriptive statistics to summarize the data, including measures of central tendency and measures of dispersion.

- Analyzed the data to identify trends, patterns, and insights that could be used to inform business decisions related to Healthcare Organizations.

VISUALIZATIONS

KPI – 1 Number of Patients across various summaries

Total Number of patients in various summary Vs states



Chain Org with higher No of Patients across various summaries

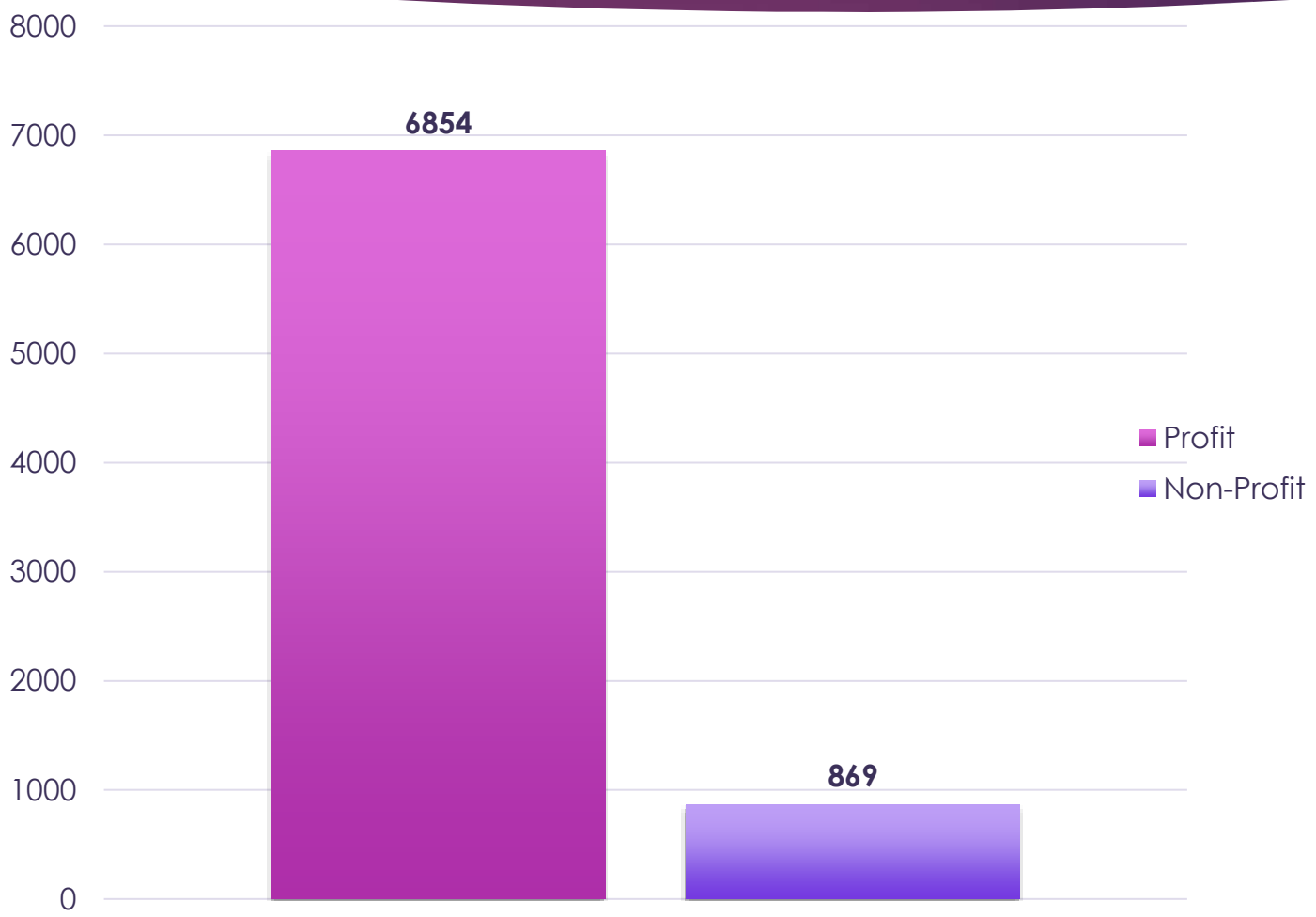
Chain Organization with higher No of Patients in various summaries

INSIGHTS

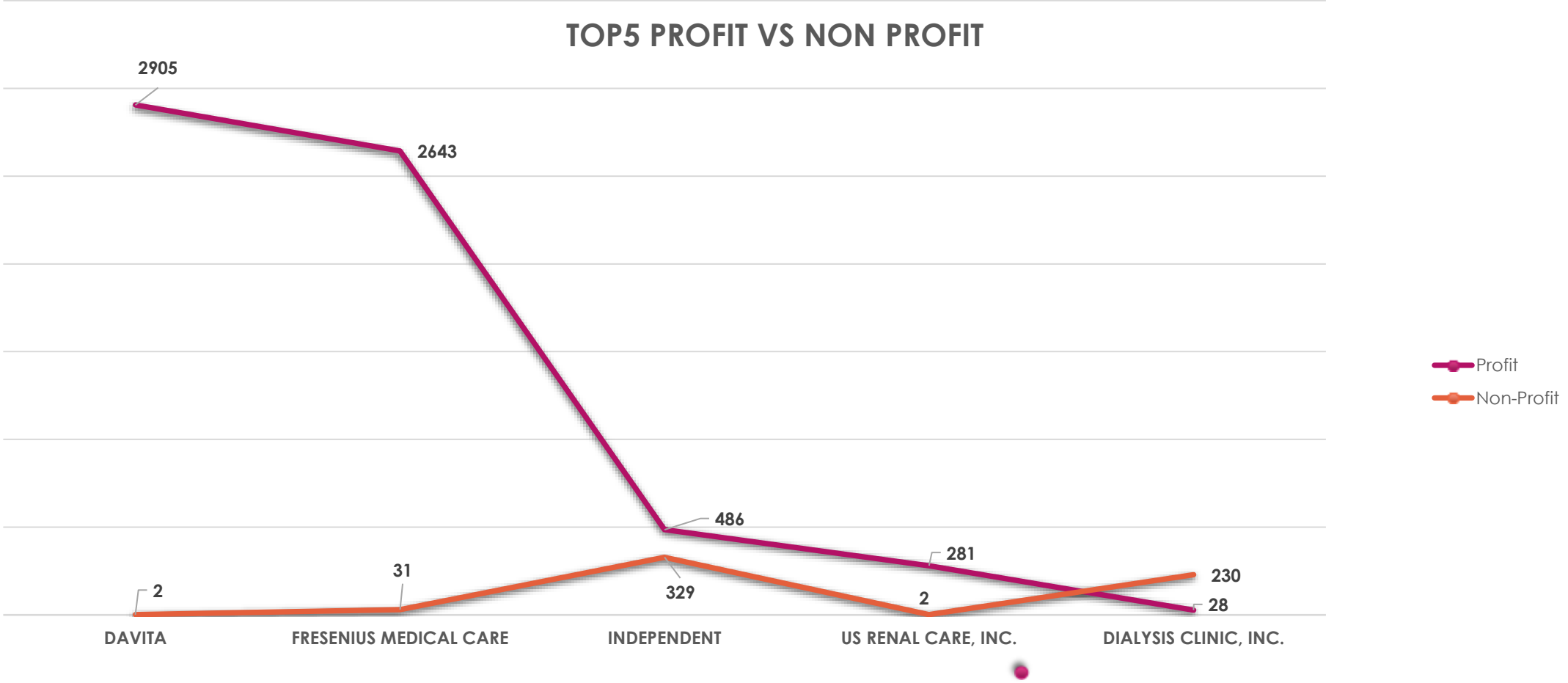
KPI – 1 Number of Patients across various summaries

- Chain Organizations with Higher Number of Patients in Various summaries have more No of Dialysis Stations.
- California, Texas, Florida and New York are the States with Highest number of patients and so more Healthcare facilities needed.
- Survival Summary shows a supremacy in total number of patients and nPCR summary is the least.
- According to Patients Hospitalization Summary Satellite Healthcare Central States LLC Is the top Profitable Facility.

KPI-2 PROFIT VS NON-PROFIT



KPI-2 TOP5 CHAIN ORG OF PROFIT VS NON-PROFIT

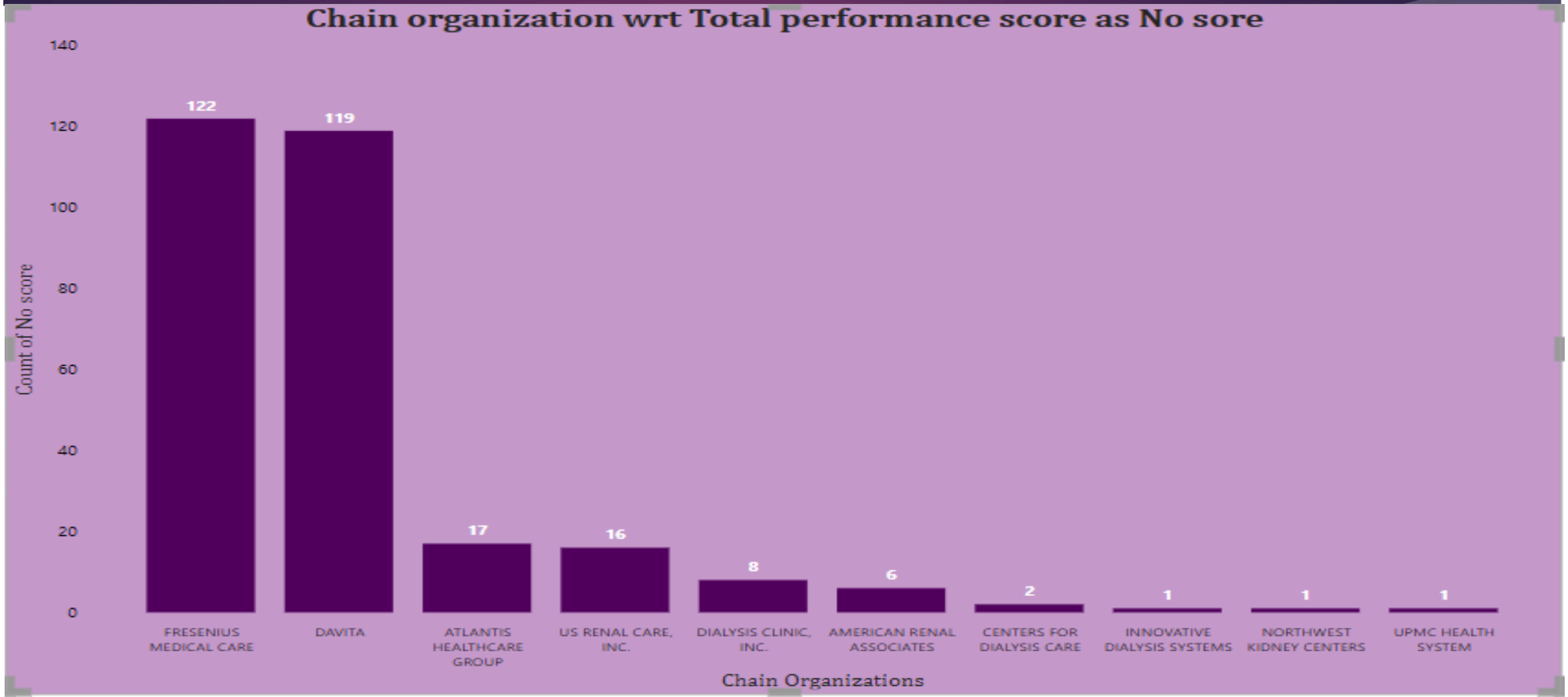


INSIGHTS

KPI – 2 PROFIT VS NON-PROFIT

- Total Count of Profit is 6854 and for Non-Profit is 869.
- Top 5 Combination of both Profit and Non-profit with respect to Chain Organization is Highest for Davita with 2907 while Least is for Dialysis Clinic Chain Org with 258.
- Main reason for those Chain Organization who has much more Profits is because of High Quality Treatments, also provides high class services to their patients.

KPI-3 Chain Organizations w.r.t. Total Performance Score as No Score

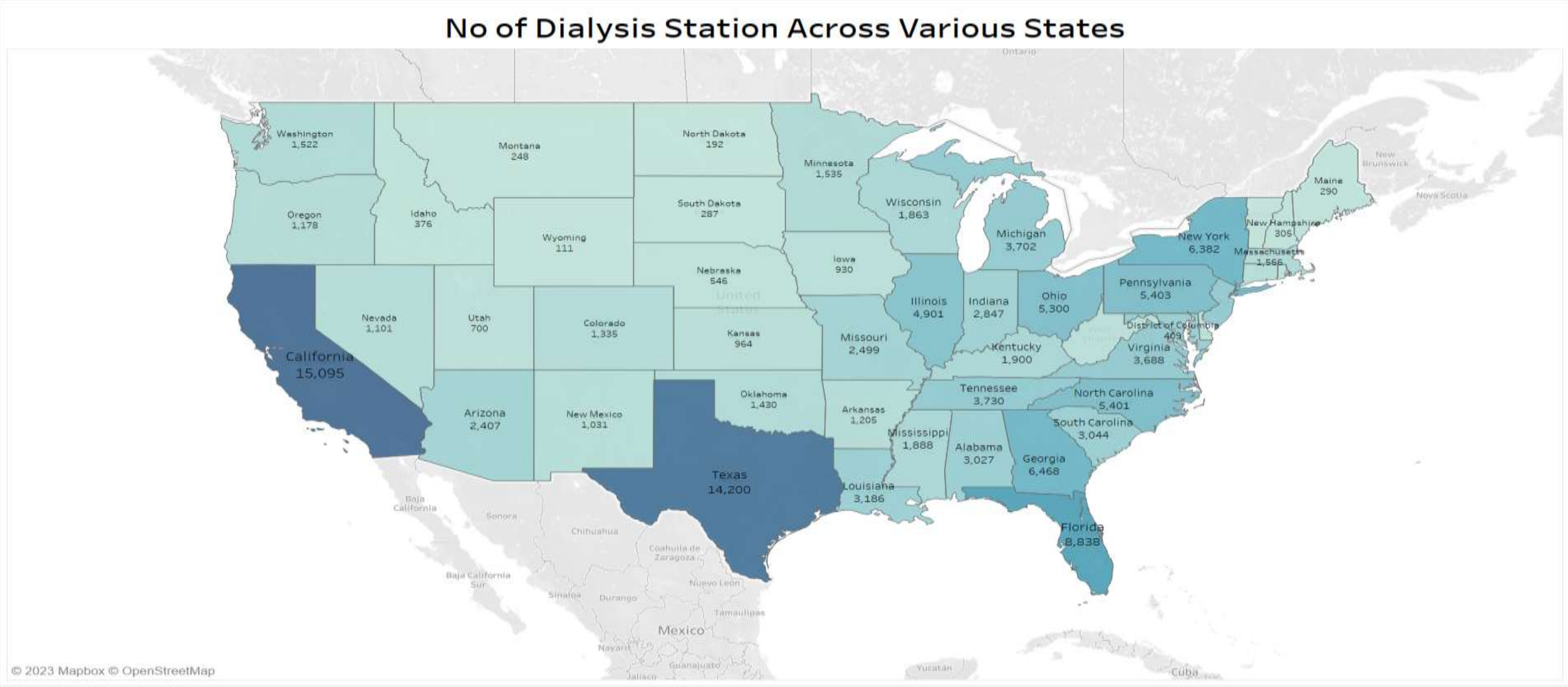


INSIGHTS

KPI - 3 Chain Organizations w.r.t. Total Performance Score as No Score

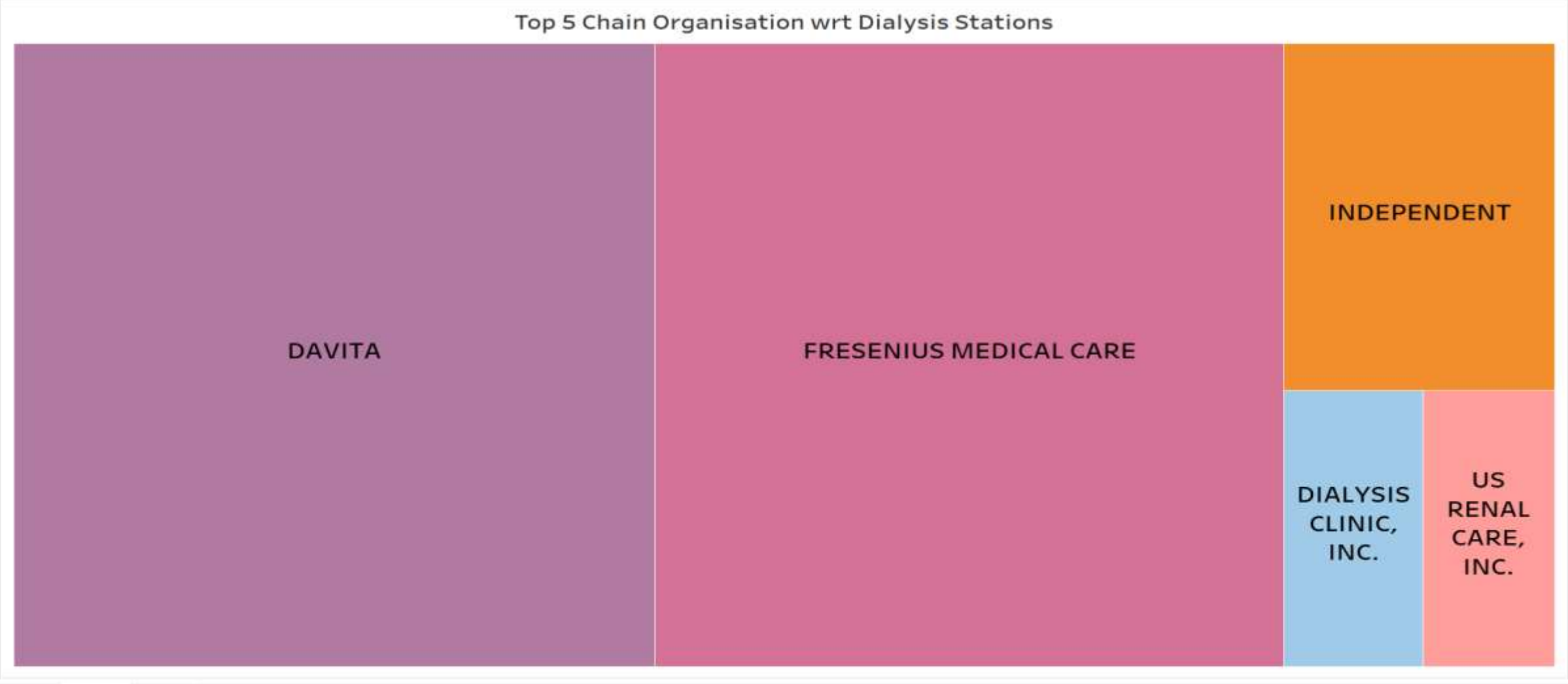
- The chain organizations to strive for no score as total performance score shows a five star rating indicates that the chain organizations Fresenius Medical Care , Davita .. are performing well across various metrics such as patient satisfaction, clinical quality and financial performance .
- The average of total performance score shows that Atlantis HealthCare Group, Belmont Court Dialysis Centre has not that good performance across various metrics. So needs improvement.

KPI-4 No of dialysis station

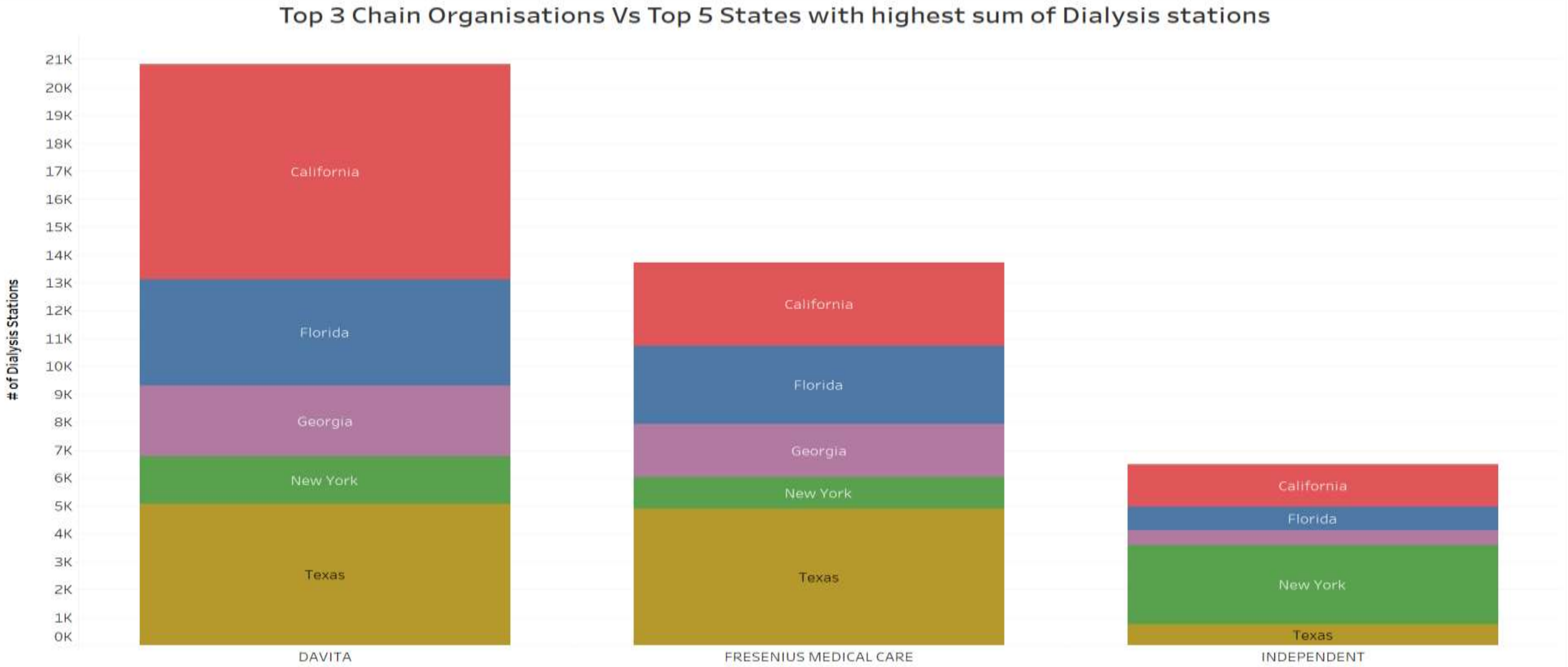


Sheet 1

KPI-4 TOP-5 CHAIN ORGANISATION WRT DIALYSIS STATION



KPI-4 TOP 3 CHAIN ORGANISATION VS TOP 5 STATES WITH HIGHEST SUM OF DIALYSIS STATION



INSIGHTS

KPI – 4 DIALYSIS STATION STATS.

•States with more Dialysis Stations have better healthcare and higher earning capacity.

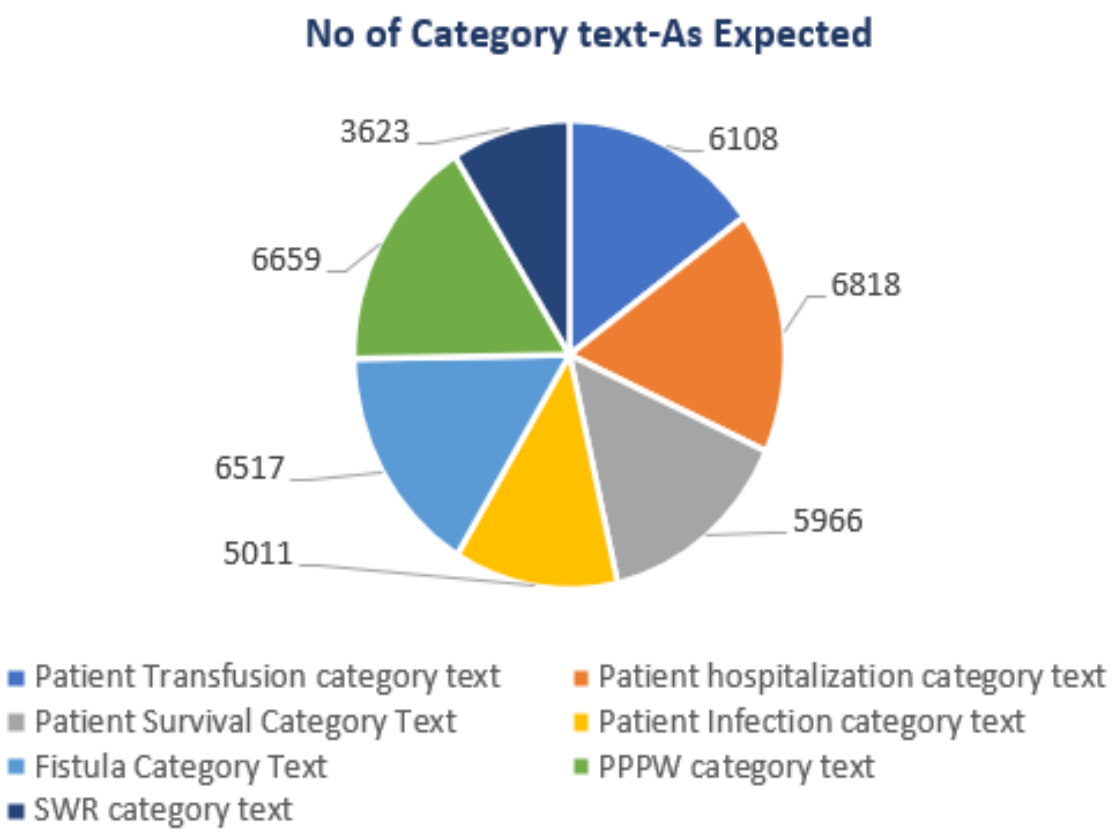
•California, Texas, New York, Florida and Georgia are the top-performing states for Dialysis Stations.

•Coastal states have higher spending capacity and better healthcare than inland states.

•Davita and Fresenius Medical care are the dominant chain organization with a combined market share of 73%.

•Davita and Fresenius have a significant presence in the top 5 states with the most Dialysis Stations, while other organizations operate locally.

KPI-5 No of category text-As Expect



INSIGHTS

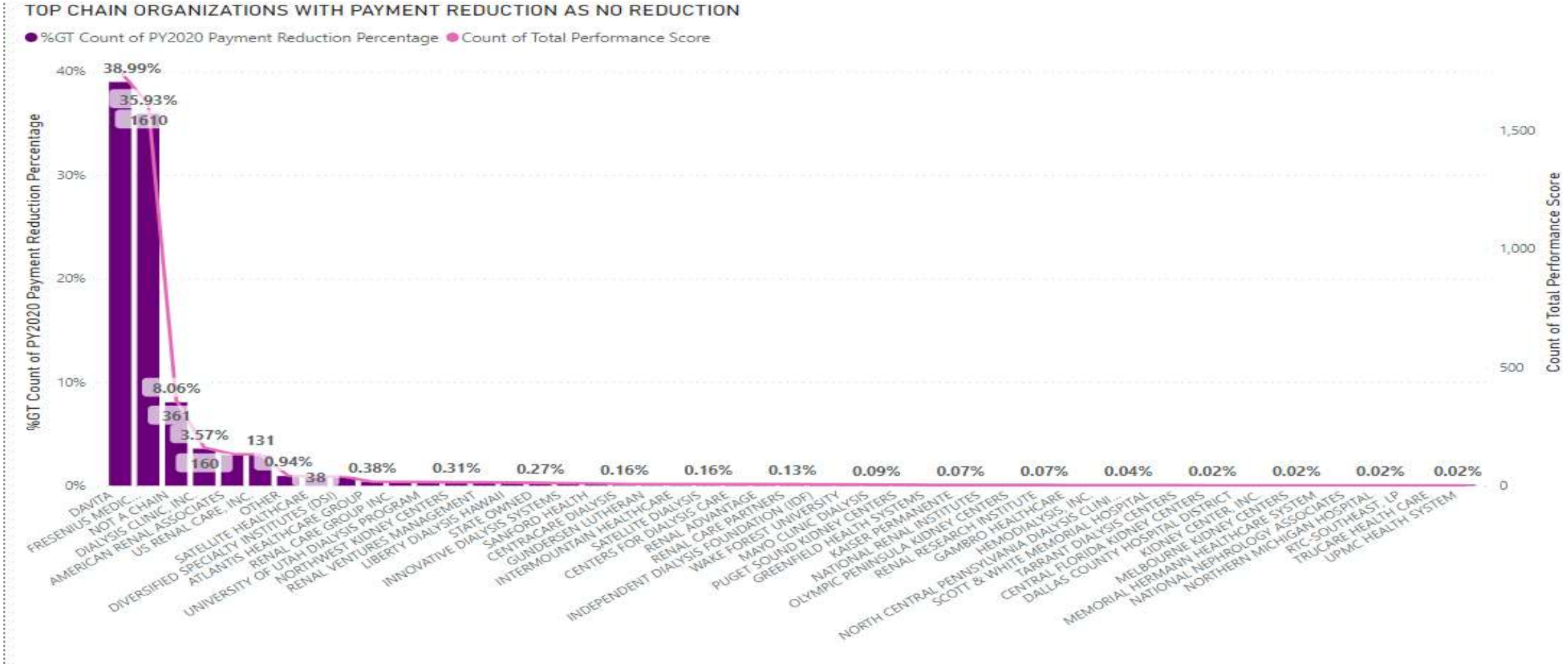
KPI – 5 NO OF CATEGORY TEXT AS EXPECTED

- We have Total 7 fields of Category text as Expected.
- Patient Hospitalization Category text has a more expected values with 6818.
- SWR Category text is the less Expected values with 3623.
- Davita, Fresenius medical care are the chain organizations which shows a more count for all the seven category of text.
- Texas, California, Florida…are the states which shows more as expected for patient hospitalization category text.

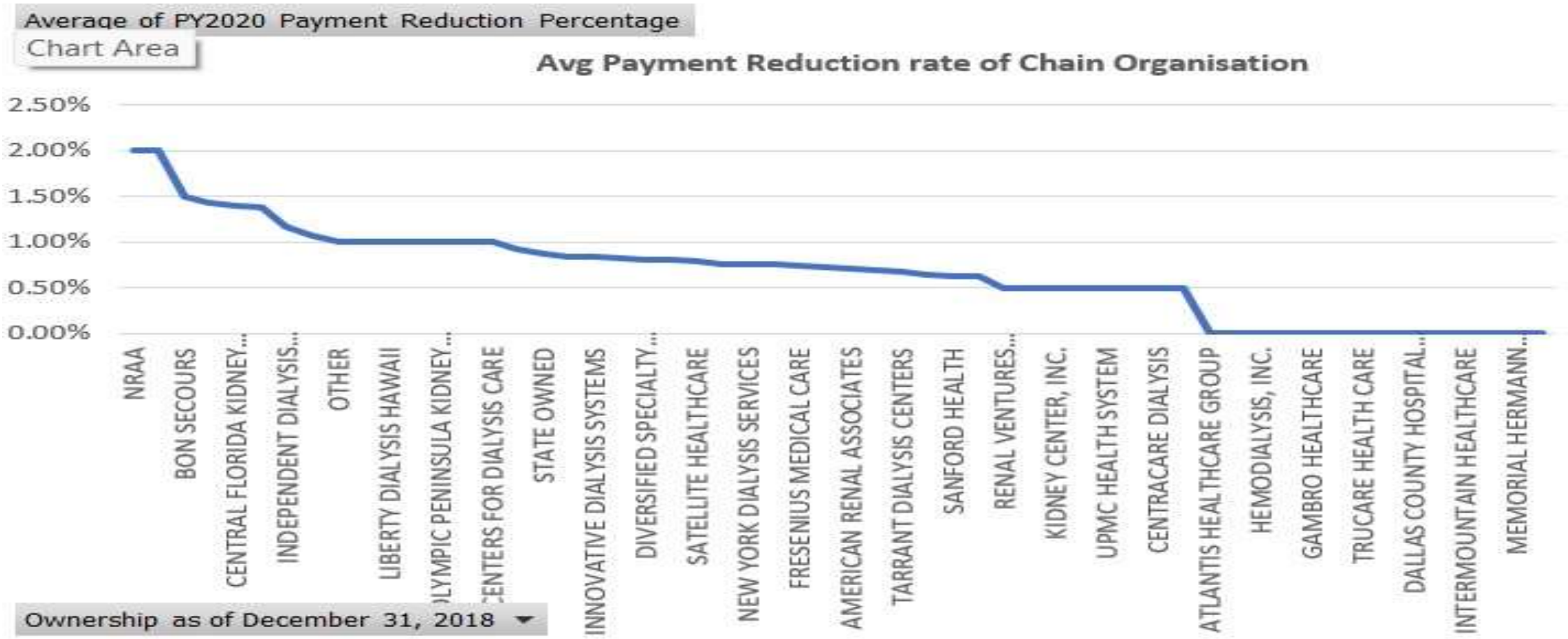
KPI – 6 AVG PAYMENT REDUCTION RATE

- The payment reduction rate is a measure of how much payments have been reduced on average for chain organizations. The total Average payment reduction is 0.81%. Which indicates that 81% payment is reduced that is, maybe the Dialysis stations operates approximately on 20% profit.

KPI – 6 PAYMENT REDUCTION RATE AS NO REDUCTION



KPI-6 AVG PAYMENT REDCUTION RATE OF CHAIN ORG



INSIGHTS

KPI – 6 PAYMENT REDUCTION RATE FOR NO REDUCTION

- Till December 31st 2018 the dialysis stations which comes under ownership of chain organizations Davita, Fresenius medical care, Dialysis clinic INC , etc has no reduction for payment percentage. Hence these chain organizations have high performance score. we can conclude that these are the leading chain organizations.
- Texas, California, Florida, Georgia.. are the States in United states which contain more Facilities and has more payment reduction as no reduction.

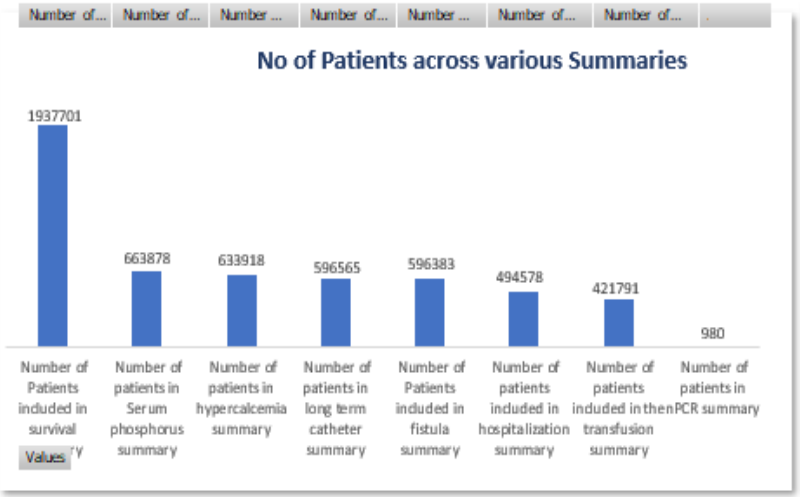
INSIGHTS

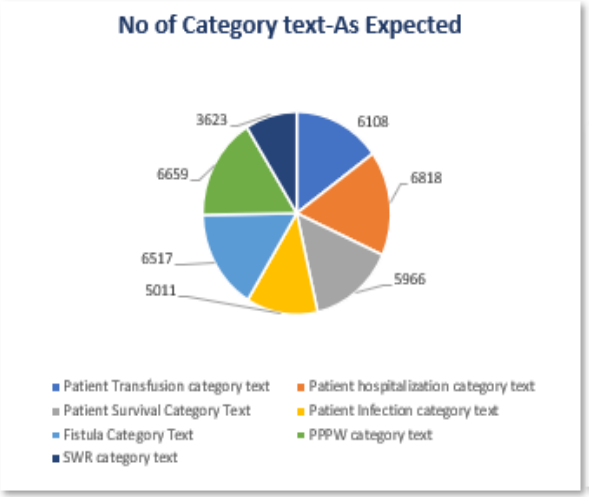
KPI – 6 AVG PAYMENT REDUCTION OF CHAIN ORG

- If a chain organization fails to meet or exceed the total performance score, it may receive a payment reduction of up to 2%.
- Here NRAA & Belmont court dialysis centre INC have average payment reduction 2% implies they fails to meet their total performance score. This indicates that these chain organization has to focus on various metrics to enhance performance score.

EXCEL DASHBOARD

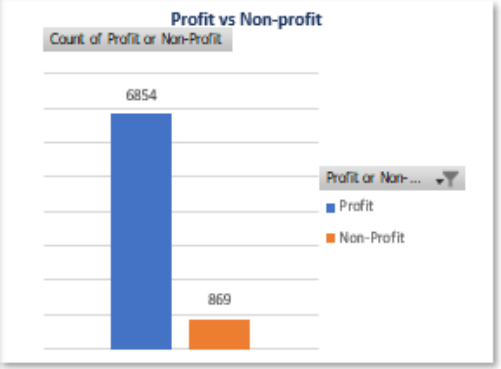
Healthcare Dashboard

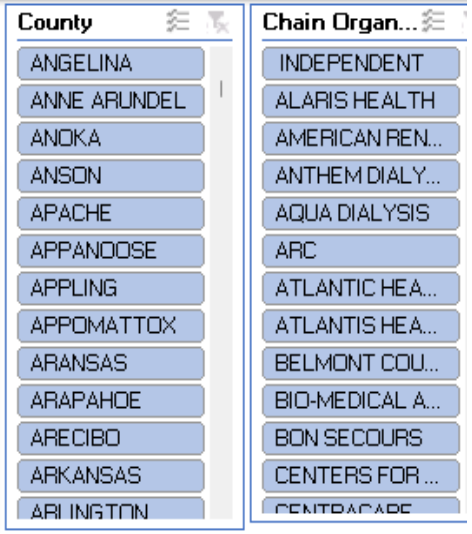


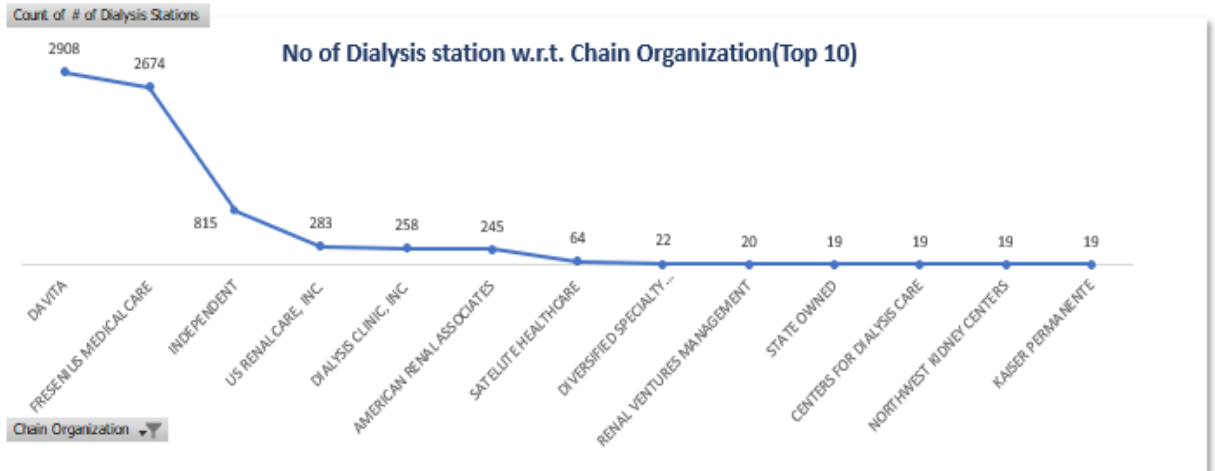


AVG Payment Reduction rate

0.81%







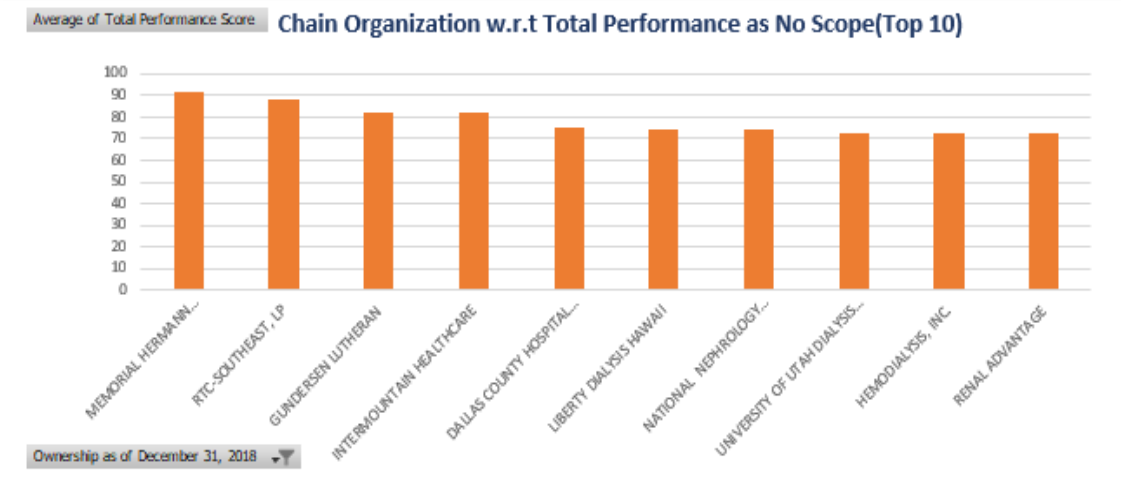
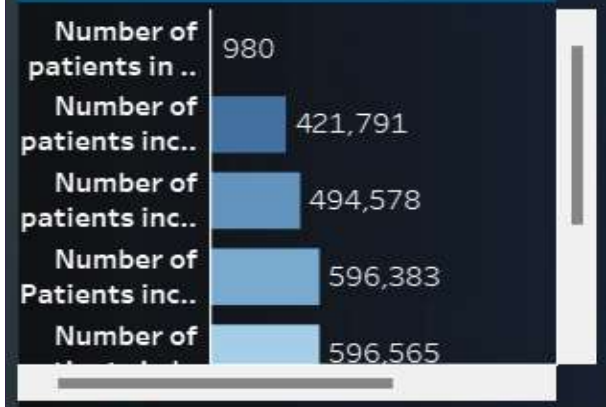


TABLEAU DASHBOARD

HEALTH CARE ANALYTICS DASHBOARD

KPI1:NUMBER OF PATIENTS ACROSS VARIOUS SUMMARY



KPI2:PROFIT VS NON-PROFIT

KPI3:CHAIN ORGANIZATION W.R.T TOTAL PERFORMANCE SCORE AS NO SCORE

Chain Organi..

NO...
NO...
NRAA
OAK...
OH...
OLY...
OU ...
PHY...
PHY...
PHY...
PHY...
PHY...
PRA...
PUG...
PUR...
PUR...
RED...
RED...
REL...
REL...
REN...
REN...
REN...
REN...
REN...

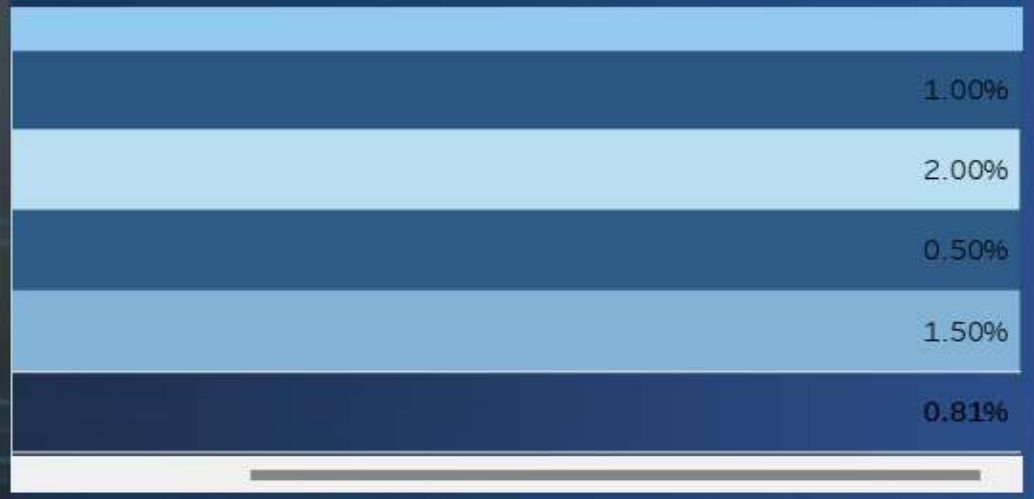
KPI4: NUMBER OF DIALYSIS STATIONS W.R.T CHAIN ORGANIZATION.

Chain Organization	
DAVITA	50,102
FRESENIUS MEDICAL CA..	49,135
INDEPENDENT	11,739
DIALYSIS CLINIC, INC.	4,822
AMERICAN RENAL ASSO..	4,608

KPI5:COUNT OF 7 CATEGORY TEXT AS EXPECTED

hospitalization category text	6,818
PPPW Category text	6,659
Fistula category text	6,517
transfusion category text	6,108
patient survival	

KPI6:AVG PAYMENT REDUCTION RATE



POWER BI DASHBOARD

HEALTH CARE ANALYTICS DASHBOARD

6818
Count of Patie...

6659
Count of PPP...

6517
Count of Fistul...

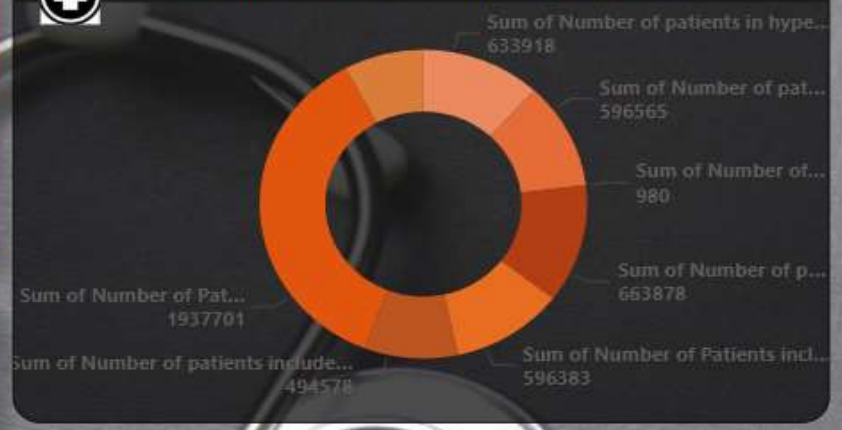
6108
Count of Patie...

5966
Count of Patie...

5011
Count of Patie...

3623
Count of SWR ...

Patients Across Various Summaries



Count of Profit or Non-Profit

AVG Payment Reduction rate

0.81%

Profit or Non-Profit

All

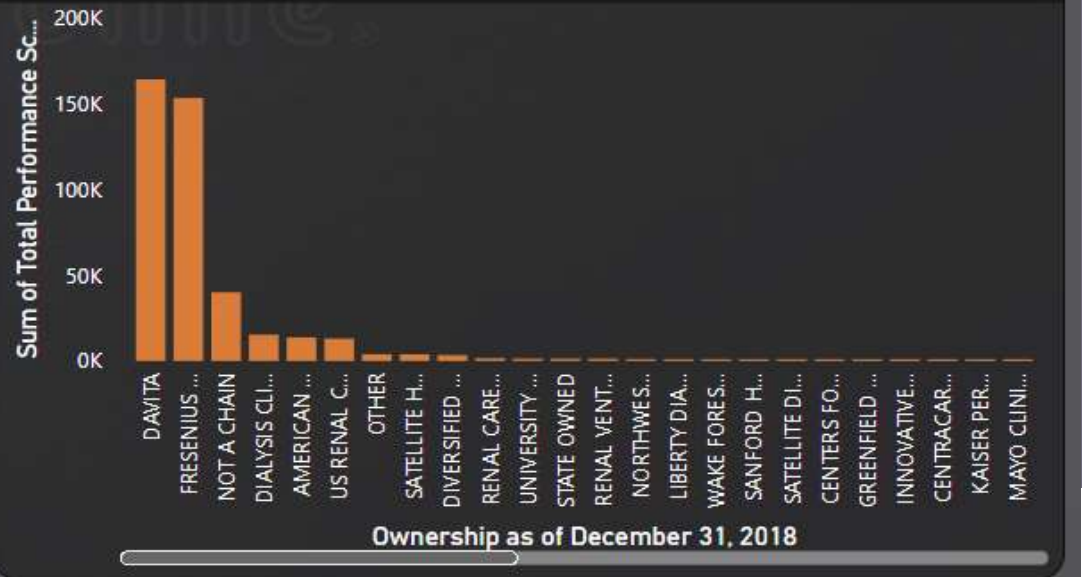
Chain Organization

- INDEPENDENT
- ALARIS HEALTH
- AMERICAN RENAL ASSO...
- ANTHEM DIALYSIS
- AQUA DIALYSIS

Number of Dialysis stations w.r.t Chain Organization

Chain Organization	Sum of # of Dialysis Stations
INDEPENDENT	37
ALARIS HEALTH	18
AMERICAN RENAL ASSOCIATES	4608
ANTHEM DIALYSIS	16
AQUA DIALYSIS	12
ARC	12
ATLANTIC HEALTH SYSTEM	30
ATLANTIS HEALTHCARE GROUP	482
BELMONT COURT DIALYSIS CENTER, INC.	22
BIO-MEDICAL APPLICATIONS OF NEVADA, INC	32
BON SECOURS	35

Sum of Total Performance Score by Ownership as of December 31, 2018



Summary

- KPI 1: Patients: California, Texas, Florida, and New York have the highest number of patients, and Chain Organizations with more patients have more Dialysis Stations.
- KPI 2: Profit vs Non-Profit: Davita has the highest number of both Profit and Non-profit chains, and organizations with more profits provide higher quality treatments and services.
- KPI 3: Total Performance Score: Fresenius Medical Care, Davita are the leading chain organizations with a five-star rating in patient satisfaction, clinical quality, and financial performance, while Atlantis HealthCare Group and Belmont Court Dialysis Centre need improvement.

Summary

- KPI 4: Dialysis Stations: California, Texas, New York, Florida, and Georgia are the top-performing states for Dialysis Stations, and Davita and Fresenius Medical Care are the dominant chain organizations.
- KPI 5: Category Text: Patient Hospitalization Category Text has the highest expected values, while SWR Category Text has the least expected values.
- KPI 6: Payment Reduction: Chain organizations with no payment reduction, such as Davita and Fresenius Medical Care, have a high performance score, while Texas, California, Florida, and Georgia have more payment reductions.

Conclusion

- The states of California, Texas, Florida, and New York have the highest number of patients and Dialysis Stations, indicating a need for more healthcare policies.
- Davita and Fresenius Medical Care are the dominant chain organizations providing high-quality treatments and services.
- Organizations with higher profits perform better, and those with no payment reductions, like Davita and Fresenius Medical Care, have a high performance score.
- Atlantis HealthCare Group and Belmont Court Dialysis Centre need improvement in their overall performance in all areas.